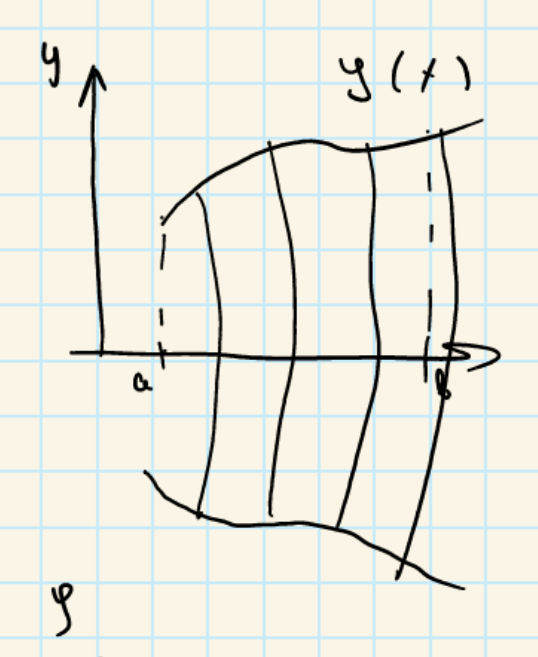
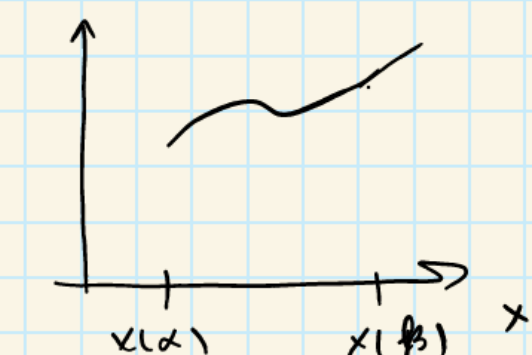
19марта(Объемы-площади)

19 марта 2026 г. 10:46

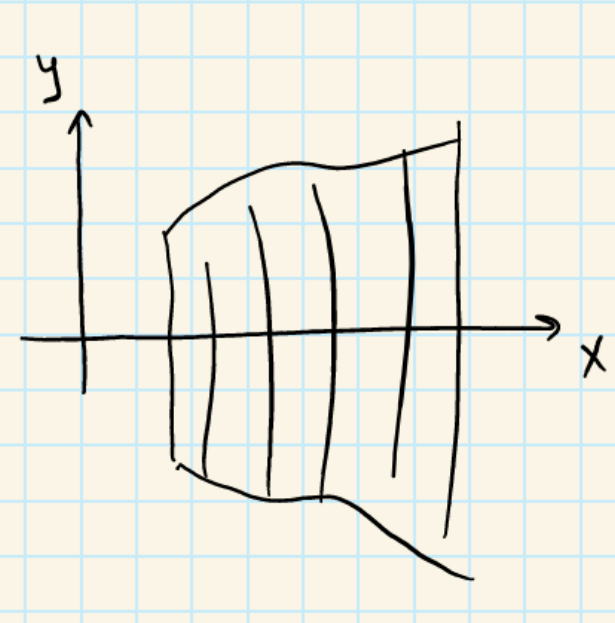


$$V = \pi \int_a^b y^2(x)\,dx$$



$y = y(t)$
$x = x(t)$

$$V = \pi \int_\alpha^\beta y^2(t)\,x'(t)\,dt$$



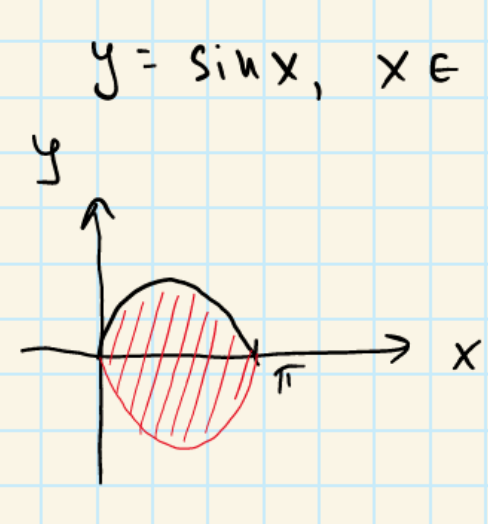
Площади:

$$S = 2\pi \int_a^b |y(x)| \sqrt{1+y'^2}\,dx$$

§8 №12.2

Объём тела вращ. вокруг оси 1) Ox
2) Oy

$y = \sin x,\ x \in 0, \pi$



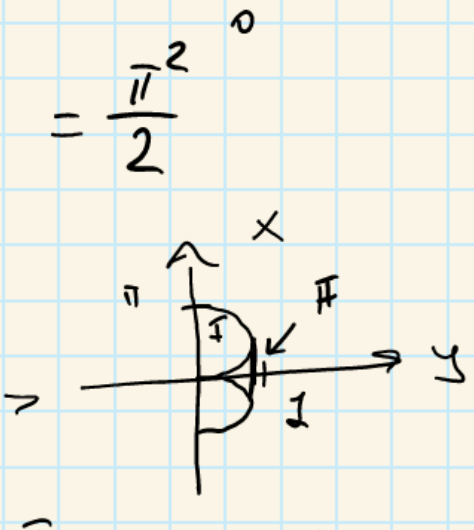
$\cos 2\alpha = 1 - 2\sin^2\alpha$

1) $V = \pi \int_0^\pi \sin^2 x\,dx = \pi \int_0^\pi \left(\frac{1-\cos 2x}{2}\right) dx$

$= \frac{\pi^2}{2}$

2)

$x \Rightarrow$



$I: \pi - \arcsin y$

$II: \arcsin y$

$V = V_I - V_{II}$

$V_{II} = \pi \int_0^1 \arcsin^2 y\,dy$

$V_I = \pi \int_0^1 (\pi - \arcsin y)^2 dy = \pi \left(\int_0^1 \pi^2 dy - 2\int_0^1 \pi \arcsin y\,dy + \int_0^1 \arcsin^2 y\,dy \right)$

$V = \pi \int_0^1 \pi^2 dy - 2\pi^2 \int_0^1 \arcsin y\,dy$

$\int_0^1 \arcsin y\,dy = \left| \begin{matrix} y = \sin t \\ dy = \cos t\,dt \end{matrix} \right| = \int_0^{\pi/2} t\cos t\,dt = t\sin t \Big|_0^{\pi/2} -$

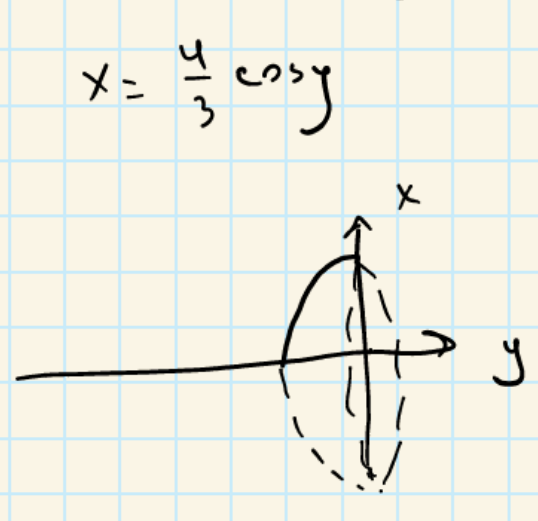
$- \int_0^{\pi/2} \sin t\,dt = \frac{\pi}{2} - 1$

Ответ: $V = 2\pi^2$

§8 №82(4)

$3x = 4\cos y,\ y \in [-\frac{\pi}{2}; 0)$ Вокруг Oy и найти S

$x = \frac{4}{3}\cos y$



$S = 2\pi \int_{-\frac{\pi}{2}}^0 \left(\frac{4}{3}\cos y\right) \sqrt{1 + \frac{16}{9}\sin^2 y}\,dy =$

$= 2\pi \int_{-\frac{\pi}{2}}^0 \frac{4}{3} \sqrt{1 + \left(\frac{4}{3}\sin y\right)^2}\,d(\sin y) =$

$= 2\pi \int_{-1}^0 \frac{4}{3}\sqrt{1 + \left(\frac{4}{3}t\right)^2}\,dt = 2\pi \int^0 \frac{4}{3}\sqrt{1 + sh^2 z}\ d\left(\frac{3}{4} ch z\right) =$

$= \ldots = 2\pi \int_{-z}^0 ch^2 z\,dz \ominus;\quad ch^2 z = \frac{e^{2z} + 2 + e^{-2z}}{4} = \frac{ch 2z + 1}{2}$

$\ominus\ \pi \int_{-z_1}^0 (ch 2z + 1)\,dz = \pi\left(z + \frac{sh 2z}{2}\right)\Big|_{-z_1}^0 =$

$= \pi z_1 - \frac{sh(-2z_1)}{2} = \pi z_1 + \frac{sh(2z_1)}{2}$

$\frac{4}{3} = \frac{e^{z_1} - e^{-z_1}}{2}$

$\frac{8}{3} e^{z_1} = e^{2z_1} - 1 \xrightarrow[\text{ур-е}]{\text{кв.}} z_1 = \ln 3$

Критерий Коши для несоб. интеграла

Пусть $f: [a, b) \to \mathbb{R}$, $a < b \le +\infty$ и интегр. на $[a, \xi]$ $\forall \xi \in (a; b)$

$\int_a^b f(x)dx$ сходится $\Longleftrightarrow \forall \varepsilon > 0\ \exists \delta > 0:\ \forall \xi_1, \xi_2 \in \dot{U}_\delta(b-0) \hookrightarrow \left| \int_{\xi_1}^{\xi_2} f(x)dx \right| < \varepsilon$

$|F(\xi_1) - F(\xi_2)| < \varepsilon$

$F(\xi) = \int_a^\xi f(x)dx$

Теорема. Если $f(x)$ и $g(x)$ опр. на $[a; b)$, $a < b \le +\infty$ и интегрируемы на $[a, \xi]$ $\forall \xi \in (a, b)$ и $f(x) \ge 0$, $g(x) \ge 0$ на $[a, b)$, тогда

1) Если $f(x) \ge g(x)$ на $[a, b)$, то
$g(x)$ - расх $\to f(x)$ расх.
$f(x)$ сход $\to g(x)$ сход.

2) Если $\lim\limits_{\xi \to b-0} \frac{f(x)}{g(x)} = k \ne 0$, то $\int_a^b f(x)\,dx$ - сходится $\Longleftrightarrow$ сходится $g(x)$

$\int_{+0}^{1/2} \frac{dx}{x^\alpha \ln^\beta x}$
$\alpha < 1$ CX
$\alpha > 1$ PX
$\alpha = 1 \to \beta > 1$ CX, $\beta \le 1$ PX

$\int_2^{+\infty} \frac{dx}{x^\alpha \ln^\beta x}$
$\alpha > 1$ CX
$\alpha < 1$ PX
$\alpha = 1 \to \beta > 1$ CX, $\beta \le 1$ PX

Лемма $\underbrace{\int_a^b f(x)dx}_{V} \overset{?}{=} \underbrace{\int_a^c f(x)dx}_{I} + \underbrace{\int_c^b f(x)dx}_{II}$

I и II сходятся => V сх.

§11 №57 $\int_0^8 \frac{dx}{x^2 + \sqrt[3]{x}}$; $\lim\limits_{x \to +0} \frac{x^2 + \sqrt[3]{x}}{\sqrt[3]{x}} = 1 \ne 0$

$\overset{cx}{\sim} \int_{+0}^8 \frac{dx}{\sqrt[3]{x}}$ = сх - ся

$\int_0^1 \frac{dx}{\sqrt[5]{1 - x^{10}}} = -\int_1^0 \frac{dy}{\sqrt[5]{1 - (1-y)^{10}}} = \int_0^1 \frac{dy}{\sqrt[5]{1 - (1-y)^{10}}}$

$y = 1 - x$

$\lim\limits_{y \to 0} \frac{\sqrt[5]{1 - (1-y)^{10}}}{\sqrt[5]{y}} = \lim\limits_{y \to 0} \frac{\sqrt[5]{1 - 1 + 10y + o(y)}}{\sqrt[5]{y}} =$

$= \lim\limits_{y \to 0} \frac{\sqrt[5]{10y + o(y)}}{\sqrt[5]{y}} = \lim\limits_{y \to 0} \frac{\sqrt[5]{10 + o(1)}}{1} = \sqrt[5]{10} + o(1) \ne 0$

Тогда интегралы экв. по сх-ти:

$\int_0^1 \frac{dy}{\sqrt[5]{y}}$ - сх-ся. $\alpha = 1/5$, $\beta = 0$

§11 №62

$\int_0^\pi \frac{dx}{\sqrt[3]{\sin x}} \overset{?}{=} \int_0^{\pi/2} \frac{dx}{\sqrt[3]{\sin x}} + \int_{\pi/2}^\pi \frac{dx}{\ldots}$

$\int_{+0}^{\pi/2} \frac{dx}{\sqrt[3]{\sin x}}$ $\lim\limits_{x \to 0+0} \frac{\sqrt[3]{\sin x}}{\sqrt[3]{x}} = 1 \Rightarrow \overset{cx}{\sim}$

$\int_0^{\pi/2} \frac{dx}{\sqrt[3]{x}}$ сходится.

$\int_{\pi/2}^\pi \frac{dx}{\sqrt[3]{\sin x}} = -\int_{\pi/2}^0 \frac{dy}{\sqrt[3]{\sin(\pi - y)}} = \int_0^{\pi/2} \frac{dy}{\sqrt[3]{\sin y}}$ - CX

§11 №72 $\int_0^\pi \frac{\ln x}{\sqrt{\sin x}}dx \overset{?}{=} \underbrace{\int_0^{\pi/2} \frac{\ln x}{\sqrt{\sin x}}dx}_{I} + \underbrace{\int_{\pi/2}^\pi \frac{\ln x\,dx}{\sqrt{\sin x}}}_{II}$

I) $\int_0^{\pi/2} \frac{\ln x}{\sqrt{\sin x}}dx \overset{экв}{\sim} \int_0^{\pi/2} \frac{\ln x}{\sqrt{x}}dx$ $\alpha = 1/2$, $\beta = -1$
сходится.

$\lim\limits_{x \to 0} \frac{\sqrt{\sin x}}{\sqrt{x}} = 1$

$\int_{\pi/2}^\pi \frac{\ln x\,dx}{\sqrt{\sin x}} = -\int_{\pi/2}^0 \frac{\ln(\pi - x)dx}{\sqrt{\sin(\pi - x)}} = \int_0^{\pi/2} \frac{\ln(\pi - y)dy}{\sqrt{\sin y}}$ - CX.

Итог: ф-я сходится.

§11 №94

$\int_0^1 \frac{\operatorname{arctg}(x^2 + x^{2\alpha})}{x \cdot \ln^\alpha(1+x)}dx$

1) $\alpha \ge 1$ $\lim\limits_{x \to 0} \frac{\operatorname{arctg}(x^2 + x^{2\alpha})}{x^2} = \lim\limits_{x \to 0} \frac{x^2 + x^{2\alpha} + o(x^2)}{x^2} =$

$= 1 \in \mathbb{R}$

$\lim\limits_{x \to 0} \frac{\ln(1+x)}{x} = 1 \in \mathbb{R}$

$\int_{+0}^1 \frac{x^2}{x^{1+\alpha}} = \int_0^1 \frac{1}{x^{\alpha - 1}}dx$
$\alpha - 1 < 1$
$\alpha < 2$ - CX - ся
$\alpha \ge 2$ PX

Вывод: при $\alpha \in [1, 2)$ - CX

2) $\alpha \in [0, 1)$ $\frac{\operatorname{arctg}(x^2 + x^{2\alpha})}{x^2} \to 1$

$\sim \int_0^1 \frac{x^{2\alpha}}{x^{\alpha + 1}}dx = \int_{+0}^1 \frac{dx}{x^{1-\alpha}}$
$1 - \alpha < 1$
$0 < \alpha$ - CX
$\alpha \le 0$ - PX

Вывод: $\alpha \in (0, 1)$ - CX

3) $\alpha < 0$ $\operatorname{arctg} x^2 + x^{2\alpha} \to C$

$\sim \int_0^1 \frac{1}{x^{1-\alpha}}dx \Rightarrow$ $1 + \alpha < 1$
$\alpha < 0$

Ответ: пупупу

§11 №98 $\int_0^{\pi/2} \sin^\alpha x \cos^\beta x\,dx \overset{?}{=} \int_0^{\pi/4} \sin^\alpha x \cos^\beta x\,dx +$

$+ \int_{\pi/4}^{\pi/2} \sin^\alpha x \cos^\beta x\,dx$

$\int_0^{\pi/4} \sin^\alpha x \cos^\beta x\,dx \overset{cx}{\sim} \int_0^{\pi/4} \sin^\alpha x\,dx \overset{cx}{\sim} \int_0^{\pi/4} x^\alpha dx$ $\alpha > -1$ CX
$\alpha \le -1$ PX

$\int_{\pi/4}^{\pi/2} \sin^\alpha x \cos^\beta x\,dx \overset{cx}{\sim} \int_{\pi/4}^{\pi/2} \cos^\beta x\,dx \ominus$

$\ominus \int_0^{\pi/4} \cos^\beta\left(\frac{\pi}{2} - y\right)dy = \int_0^{\pi/4} \sin^\beta y\,dy \overset{cx}{\sim} \int_0^{\pi/4} y^\beta dy$ $\beta > -1$ CX

Если $\alpha > -1$, $\beta > -1$ и сх, то сумма CX